

Feature Category	Feature Name	Formula / Calculation Method	Biological Meaning of a LOW Score	Biological Meaning of a HIGH Score
Morphological	Area	Sum of pixels within the nucleus boundary.	A small nucleus (e.g., lymphocyte, quiescent cell).	A large nucleus (e.g., fibroblast, cell in G2/M phase).
Morphological	Perimeter	Length of the nucleus boundary in pixels.	A small nucleus with a smooth, simple boundary.	A large nucleus or one with an irregular, complex boundary.
Morphological	Solidity	Area / Convex Area	A nucleus with significant indentations or concavity (lobulated).	A solid, convex nucleus with no major indentations.
Morphological	Extent	Area / Bounding Box Area	An irregular or elongated shape that doesn't fill its bounding box.	A squarish or rectangular nucleus.
Morphological	Eccentricity	How much the nucleus deviates from a perfect circle (0=circle, 1=line).	A highly circular nucleus.	A highly elongated, "cigar-shaped" nucleus (e.g., fibroblast).
Morphological	Major/Minor Axis	Length of the longest and shortest axes of the best-fit ellipse.	A small nucleus.	A large nucleus.
Morphological	Circularity	$(4 * \pi * \text{Area}) / (\text{Perimeter}^2)$	An irregular or elongated nucleus.	A nucleus that is almost a perfect circle.
Morphological	Orientation	Angle of the major axis relative to the horizontal axis.	(Population-level) Indicates alignment along one axis.	(Population-level) Indicates alignment along another axis.
Intensity	Mean/Max/Min Intensity	Statistical measures of pixel values within the nucleus.	A dim nucleus, suggesting diffuse, open chromatin (euchromatin).	A very bright nucleus, suggesting high DNA content or compacted chromatin.
Intensity	Intensity Std Dev	Standard deviation of pixel values within the nucleus.	A smooth, homogeneous nucleus with evenly distributed chromatin.	A heterogeneous nucleus with a mix of bright and dark patches.
Textural	GLCM Contrast	Measures local intensity variations.	A smooth nucleus with low contrast between neighboring pixels.	A coarse, "busy" nucleus with sharp transitions between bright and dark.
Textural	GLCM Homogeneity	Measures similarity of neighboring pixels.	A heterogeneous texture where neighboring pixels often differ.	A very smooth texture where adjacent pixels are similar.
Textural	GLCM Energy	Sum of squared elements in the GLCM.	A complex, disordered texture.	An orderly, uniform, or periodic texture.
Textural	GLCM Correlation	Measures the linear dependency of neighboring pixel intensities.	A random, chaotic texture.	A highly structured, linear, or repetitive texture.
Contextual	Cell Height (y)	The y-coordinate of the cell's centroid.	The cell is located near the top of the image.	The cell is located near the bottom of the image.
Contextual	Mean Dist to 5 Neighbors	Average Euclidean distance to the centroids of the 5 closest cells.	The cell is in a dense, tightly packed region.	The cell is isolated or in a very sparse region.
Sub-nuclear	N Chromocenters	Count of distinct bright, condensed spots (chromocenters) in the nucleus.	A nucleus with diffuse chromatin and few or no bright foci.	A nucleus with many distinct, punctate bright spots.
Sub-nuclear	Mean Chromocenter Area	Average size of the detected chromocenters.	The nucleus has no chromocenters, or they are very small.	The nucleus has large, prominent chromocenters.
Sub-nuclear	Chromocenter Area Fraction	Percentage of the total nuclear area occupied by chromocenters.	Chromocenters are absent or cover a very small fraction of the nucleus.	Chromocenters are a dominant feature, covering a large part of the nucleus.